

# JOHN WRIGHT

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## PROFESSIONAL SUMMARY

M.S. Computer Science graduate with hands-on project experience in C/C++, Python, TypeScript, Unreal Engine 5, automated testing, CI, systems programming, and full-stack development. Built a dedicated-server multiplayer VR prototype with replicated interactions and server-authoritative gameplay logic, a deployed full-stack application with automated regression testing and GitHub Actions CI, and a concurrent C client-server application. Seeking software engineering, QA automation, Unreal Engine, VR/XR, and simulation roles.

## TECHNICAL PROJECT EXPERIENCE

### Independent Software Development

*Grand Junction, CO | May 2023 – Present*

- **Multiplayer VR Prototype:** Built a dedicated-server multiplayer VR prototype in Unreal Engine 5 using Blueprints and the VR Expansion Plugin (VRE). Integrated VRE's physics-based gripping and held-object interactions, implemented replicated flashlight state, server-authoritative melee damage, enemy health and ragdoll death behavior, and PawnSensing-driven combat AI.
- **Top-Down GAS RPG:** Developed a top-down Unreal Engine 5 RPG project using C++, Blueprints, and the Gameplay Ability System. Implemented enemy combat, experience progression, save-slot persistence, ability loadouts, and upgradeable fireball, lightning, and arcane-shard abilities.
- **Full-Stack Task Manager:** Built and deployed a T3-stack TypeScript task-management application using Next.js, tRPC, Prisma, PostgreSQL/Neon, Tailwind CSS, and Auth.js email/password authentication with email verification. Implemented server-side task ownership and authorization through protected tRPC procedures, Prisma foreign keys, and user-scoped database queries.
- **Automated Testing & CI:** Created Playwright end-to-end and Python/Pytest API test suites covering login, task creation and validation, and rejection of unauthenticated API access. Configured GitHub Actions to run backend and browser regression tests against isolated Prisma/PostgreSQL schemas on pushes and pull requests.
- **Multi-Client Server:** Developed a concurrent client-server application in C using POSIX threads, UNIX domain sockets, mutex synchronization, signal handling, and thread lifecycle management.

### Senior Capstone Project – Body Farm Database

*Colorado Mesa University | Grand Junction, CO | January 2018 – May 2018*

- Gathered requirements from researchers, translated operational needs into normalized relational database schemas, created tables, relationships, ER diagrams, data dictionaries, forms, reports, and queries to support data entry and reporting.

## PROFESSIONAL EXPERIENCE

### **Independent Fabricator** | Self-Employed | Dec. 2023 – May 2025

- Designed and fabricated original welded metalwork from concept through finished pieces.
- Managed material selection, fabrication workflow, finishing, photography, product listings, and fulfillment.
- Maintained an online storefront for original work.

### **Bookkeeper** | Legacy Behavioral Services LLC | Grand Junction, CO | May 2023 – Aug. 2023

- Entered and organized financial transaction records, invoices, receipts, and expense documentation.
- Maintained accurate digital files to support bookkeeping and reporting processes.

### **Welder / Field Fabricator** | Circle P Welding | Norman, OK | May 2020 – Jan 2021

- Fabricated, fit, and welded structural steel components for municipal water-tower and water-storage-tank construction projects.

### **Audit Staff** | Dalby Wendland | Grand Junction, CO

*July 2019 – May 2020*

- Prepared audit workpapers, analyzed financial data using advanced Excel, and communicated with clients to obtain and clarify supporting documentation.

## EDUCATION

### **M.S., Computer Science** — Stevens Institute of Technology, Hoboken, NJ

*May 2025 | GPA: 3.934*

### **B.S., Computer Information Systems; B.S., Public Accounting** — Colorado Mesa University, Grand Junction, CO

*December 2018 | GPA: 3.76*

## SKILLS

**Languages:** Python, TypeScript, JavaScript, C++, C, SQL

**Unreal Engine:** Unreal Engine 5, Blueprints, C++, VR Expansion Plugin, Gameplay Ability System, Multiplayer Replication

**Testing & Automation:** Playwright, Selenium, Pytest, Python Requests, Postman, UI Testing, API Testing, Regression Testing, Page Object Model

**Web & Backend:** Next.js, React, tRPC, Prisma, Tailwind CSS, PostgreSQL, Authentication, HTTP APIs

**Tools & Deployment:** Git, GitHub, GitHub Actions, Diversion, Vercel, Neon

